

Chapter 12

AI and Strategy: The Three Perspectives Framework

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Abstract

This chapter introduces a generic framework that elucidates the intersection of Artificial Intelligence (AI) and Business Strategy. Despite the growing importance of AI technology in modern business models, there is still a lack of comprehensive understanding of its strategic implications in academic research. Through a literature review and analysis of practical cases, this study delves into the evolving landscape where AI and Strategy intersect, highlighting its crucial role across various business sectors. Based on our analysis, we present the Three Perspectives of AI and Strategy Framework: *AI for Strategy*, *AI in Strategy*, and *AI as Strategy*. These perspectives categorize how AI is integrated into business operations. This framework provides a comprehensive understanding of AI's strategic significance, offering organizations guidance on leveraging AI to gain a competitive edge. Importantly, it assists organizations in decision-making regarding the adoption of these integrations within their specific business contexts.

Keywords: Artificial intelligence (AI); business strategy; generative AI; business model; technology

1. Introduction

This chapter introduces a novel framework to understand the role of Artificial Intelligence (AI) in strategic business planning and decision-making. We showcase the intersection of AI and strategy, shedding light on the evolving landscape where these two domains converge. As AI technology rapidly evolves, its integration into business strategy becomes increasingly complex and essential. The literature review

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by [Borges et al. \(2021\)](#) shows the prevalent application of AI in relation to business strategies. Despite its significant potential, the study indicates that the intersection in question has not been thoroughly studied in academic research.

AI technology has evolved beyond the speculative visions of futurists and has become a fundamental component in the business models of many organizations. It plays a vital strategic role in the plans of diverse business sectors. Despite this, it is evident that there will be winners and losers, necessitating decision-makers to be strategic for the future ([Dwivedi et al., 2021](#)). Building on this imperative, MIT Sloan Management Review conducted an exhaustive survey on a global scale, involving over 3,000 executives, managers, and analysts from diverse industries. Furthermore, the research incorporated extensive interviews with more than 30 technology experts and executives. The prevailing perspective among organizations is that AI is regarded as a strategic opportunity, acknowledged as a pivotal pathway to attain a competitive advantage ([Ransbotham et al., 2017](#)). For example, according to literature review conducted by [Borges et al. \(2021\)](#), the strategic application of AI aids decision-making, improves customer relationship through automation, engagement of customers, and employees; and enables machine-to-machine communication during new products and services. Notwithstanding these contributions, a paucity of knowledge persists in our understanding of AI and strategy intersection. Therefore, in an era where AI increasingly reshapes business landscapes, understanding its strategic implications is paramount for organizations.

Moreover, our objective is not merely to conduct a literature review of AI strategy or Information Technology-based strategy, rather, we aim to explore the insights provided by existing literature regarding the intersection of AI and Strategy, along with how organizations are capitalizing on this intersection. This study serves to address the existing research gap in the area and offers the latest overview of the literature. Additionally, it is important to underscore that this review aims to conceptualize the intersection of AI and Strategy, offering a generic framework delineating its landscape.

As a result of our literature review, in this chapter, we introduce ‘The Three Perspectives of AI and Strategy Framework’, which presents an aligning approach to conceptualize AI’s role in business strategy. This framework categorizes AI’s integration into business practices into three distinct perspectives: *AI as Strategy*, *AI in Strategy*, and *AI for Strategy*. Each represents a unique approach to leveraging AI – from being the core of business strategy to a tool within strategic planning to a facilitator of strategic decision-making. This comprehensive view addresses the growing complexity of AI in the modern business world, offering a lens to view how organizations have integrated AI into their strategy for gaining competitive advantages, fostering strategic decision-making, and driving sustainable growth. As AI continues to evolve, this framework serves as a guide for both viewing the positions of competing organizations’ strategies and to reflect one’s strategy concerning the use of AI.

2. Evolution of Strategy and AI: Contextualizing the Field

Tracing the evolution of AI within the business strategy landscape shows the increasing intertwining of traditional business models with technological advancements. The integration of AI in business strategy marks a pivotal shift in the corporate world (Davenport & Ronanki, 2018). Historically, businesses leveraged technology as a tool to enhance operations, but the advent of AI has catalyzed a more profound transformation (McAfee & Brynjolfsson, 2017). AI's capabilities have evolved from simple automation to complex decision-making, influencing both operational and strategic realms (Fountaine et al., 2019). This progression necessitates a fresh perspective on AI's role in strategic planning. Traditional business strategies, primarily centred around human decision-making and market analysis, are now being re-evaluated and restructured to incorporate AI's data-driven insights. The 'Three Perspectives of AI and Strategy Framework' emerges from this evolution, providing a contemporary lens to view the position of an organization with AI in the strategic context. It reflects the changing dynamics where AI is not just an operational tool but a strategic asset, a source of innovation, and a key driver in decision-making processes. This background sets the stage for understanding the significance and application of this framework in today's business environment.

Viewing from strategic management perspective, business strategy and information technology have been recognized as intertwined components crucial for organizational success and adaptation in an ever-evolving business landscape. Organizational transformation through informational technology, encompassing structures, processes, and strategies, enabling companies to adapt and thrive in a rapidly changing environment, is not a new concept (see Morton, 1991). However, an early study by Luftman et al. (1993) show that viewing information technology and business strategy in isolation is inadequate. The study argues that strategic utilization of IT is imperative for businesses, as it holds the power to fundamentally reshape industries. Nonetheless, realizing effective and efficient IT utilization necessitates aligning information technology strategies with business strategies, a crucial aspect often neglected in conventional approaches. Moreover, Coltman et al. (2015) suggest that organizations achieve optimal performance when they align key informational technology resources with their business strategy and employ suitable structures to oversee the deployment and efficient management of these resources. However, the integration and alignment of IT and strategy pose a significant challenge without clearly defining the strategic implications and desired outcomes for the organization. Bharadwaj et al. (2013) identified four themes that guide our discussion and serve as a framework to elucidate the intricacies of digital business strategy: scope, scale, speed, and the sources of business value creation and capture.¹ Therefore, in this study, we adopt Bharadwaj et al.'s (2013, p. 472) definition of strategy, wherein organizational strategy is conceived and implemented through the strategic utilization of digital resources aimed at creating distinctive value propositions.

¹'Digital business strategy should not be positioned below business strategy but treated as business strategy itself for the digital era' (Bharadwaj et al., 2013, p. 473; Cf. Henderson & Venkatraman, 1999).

Viewing from information technology perspective in the realm of organizational context, AI has been a subject of significant interest. Early AI research in the 1960s initially focussed on decision-making processes, characterized by problems solvable through mathematical formulas (see e.g. [Buchanan & O Connell, 2006](#); [McCarthy & Hayes, 1981](#)). By the 1980s, AI was being actively incorporated into businesses, with investments aimed at developing computerized vision systems, robots, and expert systems, marking its evolution as a strategic tool for competitive organizational differentiation ([Boden, 1984](#); [Holloway, 1979](#); [Porter, 1985](#); [Russell & Norvig, 2010](#)). The turn of the millennium saw a shift in AI research, focussing more on algorithms and dealing with the challenges posed by large databases and the turn of ‘Big Data’ ([Fan et al., 2014](#); [McAfee & Brynjolfsson, 2017](#); [Russell & Norvig, 2010](#); [Zhuang et al., 2017](#)). This period was characterized by advancements in hardware and big data analytics, facilitating the identification of patterns and insights from large datasets ([Boyd & Crawford, 2012](#); [Fan et al., 2014](#)) to support among others, decision-making and customer intelligence ([Elgendy & Elragal, 2014](#)).

A significant milestone was achieved in 2016 with Google DeepMind’s AlphaGo, a deep learning-based programme designed to play the ancient game of Go. Unlike traditional AI at the time, AlphaGo was trained through human expert moves and self-play reinforcement learning, showcasing creativity and the ability to generate novel insights, culminating in its victory over world champion Lee Sedol ([Holcomb et al., 2018](#); [Silver et al., 2016, 2017](#)). The contemporary rise of AI, attributed to substantial data volumes, advanced algorithms, and improved computational hardware, has drawn significant attention from major tech companies. Firms like Google, Amazon, Microsoft, Salesforce, and IBM are now offering cloud-based infrastructure for machine learning, and democratizing access to cognitive technologies ([Davenport, 2018](#); [Marr, 2019](#); [McAfee & Brynjolfsson, 2017](#)). This proliferation of AI has not only provided tools for organizations’ transformation ([Holmström, 2022](#)) but has also been in the actual core of daily activities of the business itself. Several major corporations are now essentially run by ‘algorithmic management’, where AI can be said to be the principal coordinator of work, replacing middle-management (see e.g. [Noponen, 2019](#); [Noponen et al., 2023](#)). Examples include the Uber ride-fairing company and delivery companies such as Wolt and Foodora ([Möhlmann & Zalmanson, 2017](#); [Wood, 2021](#)).

The Cambrian explosion of Generative AI (GenAI) in 2022 initiated what can be said to be a new and accelerated phase of the evolution of the AI landscape ([Daigle, 2023](#); [Stokel-Walker & Van Noorden, 2023](#)). According to statistics from GitHub, public repositories of Generative AI projects have had 248% year-on-year growth in November 2023 ([Daigle, 2023](#)) and the Hugging Face open-source AI community has had a 376% increase in its *text-generating* models from December 2022 to December 2023. This technological surge, particularly in the realms of Large Language Models (LLMs) and Diffusion Models, has been due to groundbreaking advancements in AI’s capabilities. LLMs, exemplified by models like GPT-4, have revolutionized the way AI understands and generates human language, enabling applications ranging from automated content creation to complex problem-solving ([Bubeck et al., 2023](#)). These models have demonstrated an unprecedented ability to generate coherent, contextually relevant text, thereby opening new avenues for AI

interaction and accessibility (Glazko et al., 2023). On the other hand, Diffusion Models have dramatically advanced the field of visual content generation (Yang et al., 2023). By iteratively refining random noise into detailed images, these models have introduced a new paradigm in digital art and design, enabling the creation of highly realistic and creative visuals from textual descriptions (Eapen et al., 2023). The fusion of linguistic and visual generative capabilities marks a significant leap in AI's ability to mimic, and in some cases, augment human creativity and intelligence.

This rapid development has not only captivated the attention of technologists and businesses but has also raised crucial ethical and societal considerations about the role and impact of AI in everyday life (Weidinger et al., 2021). The pervasive impact of these technologies, particularly LLMs, has been felt across organizational functions, from coding to strategy consultancy, and impacting work productivity, and quality in unprecedented ways (Brynjolfsson et al., 2023; Dell'Acqua et al., 2023; Dencik et al., 2023; Dohmke et al., 2023). Despite these advancements, and partly due to their speed, there exists a discernible gap in the scholarly exploration concerning the integration and utilization of AI within organizational strategy.

The recent work by Ruokonen and Ritala (2024) argues that AI is increasingly gaining strategic importance within companies. For managers, the crucial inquiry revolves around precisely defining the implications of adopting an AI-first strategy and determining the type of AI strategy to pursue. Nonetheless, the initiation and approach to embracing an AI-first strategy vary among companies, indicating differing starting points in their adoption journeys. Therefore, it becomes crucial for us to explore the diverse approaches companies take in initiating and embracing AI-first strategy.

By reviewing academic literature and examining practical cases, along with exploring the various pathways and strategies adopted by different organizations, we aim to glean valuable insights into the current state of AI implementation, trends, and opportunities that can inform strategic decision-making. This comprehensive analysis will contribute to a conceptual understanding of the evolving landscape of AI adoption and its implications for organizational success.

The following literature review examines the intersection of AI and Strategy interface.

3. Methods

This chapter's main objective is to conduct a comprehensive literature review focussing on the strategic use of AI in academic literature, supported by case examples from real-world companies. As we aim to explore and conceptualize the AI and Strategy intersection, we utilized academic sources, including peer-reviewed journals, conference proceedings, and reputable academic publications, pertaining to the strategic utilization of AI. Additionally, we explored case studies and reports from various industries to incorporate real-world applications and outcomes. For this, we comprehensively analyzed to determine the organizational approach to AI strategy. This included examining mission statements, business models, and documented strategies to understand the role of AI in shaping their overall direction.

Our selection criteria for case studies relied on a diverse set of examples from various industries that were selected to bring insight into the practical realities of AI implementation. These cases were chosen based on the publicly available information, to avoid speculating how and why AI is being used. Recent industry reports were used to paint the broader landscape of developments in the field.

Following this, in our literature review, we identified patterns in how organizations with varying AI strategy perspectives demonstrated distinct approaches to leveraging AI, leading to the conceptualization of the three perspectives. Therefore, the identification of the three perspectives – *AI as Strategy*, *AI in Strategy*, and *AI for Strategy* – is grounded in a thorough review of existing literature on Strategy and AI. Key concepts were extracted from academic articles, industry reports, and relevant frameworks. An analysis was conducted to draw insights across different industries and organizational contexts. This involved identifying commonalities and distinctions in how organizations from various sectors adopt and integrate AI into their strategic frameworks.

Our literature review adhered to systematic review guidelines as outlined by Okoli (2015) and Liberati et al. (2009). We conducted the search using a single database, Scopus, chosen for its extensive coverage across multiple disciplines (Gusenbauer & Haddaway, 2020). Employing a carefully developed search string comprised of pertinent keywords aligned with our research question, we established explicit inclusion and exclusion criteria to guide the screening process. The search string was the following:

“TITLE-ABS-KEY (“Artificial Intelligence” OR “Machine Learning” OR “Deep Learning” OR “Generative Artificial Intelligence” OR “GenAI”) AND TITLE-ABS-KEY (“strateg* plan” OR “strateg* execution” OR “strateg* implementation” OR “business strateg*” OR “corporate strategy” OR “organi*ational strategy”) AND PUBYEAR > 2014 AND PUBYEAR < 2025 AND (LIMIT-TO (SRCTYPE, “j”)) AND (LIMIT-TO (DOCTYPE, “ar”)) AND (LIMIT-TO (LANGUAGE, “English”))”.

The search phrase yielded $N = 272$ records. To qualify for inclusion, articles were required to focus substantially on both business strategy and artificial intelligence. Articles that only briefly touched upon the strategic context of artificial intelligence were excluded. Upon screening titles and abstracts, both researchers independently evaluated the relevance of each article, resolving any discrepancies through discussion. Subsequently, $N = 187$ articles were excluded, leaving $N = 85$ for full-text screening.

During the full-text screening phase, each author independently assessed the remaining articles for eligibility, with conflicts resolved through consensus. Following this rigorous evaluation, $N = 69$ articles were deemed ineligible, resulting in $N = 16$ articles being included for the final systematic literature review analysis (see Fig. 12.1).

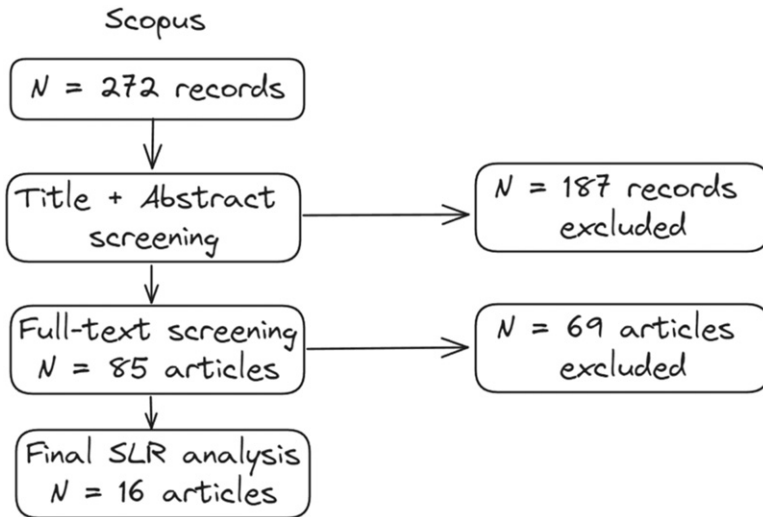


Fig. 12.1. The Flow Diagram for the Database Search of Publications for Systematic Literature Review. *Source:* Authors own work.

4. Resulted Thematic Perspectives

As a result of our literature review, we identified three overarching thematic perspectives on AI and Strategy intersection (refer [Table 12.1](#)).

Table 12.1. Literature Review Synthesis: Thematic Perspective on AI and Strategy.

Perspective	Description	Real-Life Examples	Seminal Articles
AI for Strategy	AI is used as a tool to support and enhance strategic decision-making processes. AI's analytical capabilities are leveraged to process data, uncover insights, predict trends, and inform planning. The focus	- Enel S.p.A. utilizes AI to guide strategic decisions and improve operational efficiency in the energy sector - McDonalds uses AI pilots to identify data-driven insights aligned	Kar et al. (2021) Dzhusupova et al. (2024) Gallego-Gomez and De-Pablos-Heredero (2020) Goralski and Tan (2020) Gudigantala et al. (2023) Güngör (2020)

(Continued)

Table 12.1. (Continued)

Perspective	Description	Real-Life Examples	Seminal Articles
AI in Strategy	is on augmenting human strategic thinking rather than AI driving the strategy itself.	- with business strategy - Financial institutions employ AI for market trend analysis and identifying investment opportunities - Healthcare and manufacturing industries augment strategic planning through AI-powered scenario modelling and forecasting	- Kanitz et al. (2023) - Watson et al. (2021) - Shaik et al. (2024)
	AI is deeply integrated as a core component of the organization’s overarching strategy. AI tools and capabilities are interwoven with broader strategic objectives and initiatives, often explicitly acknowledged in strategy documents. AI is crucial for achieving strategic goals but not the singular focus.	- Tesla integrates AI into its manufacturing, core products, R&D, and innovation processes as part of its comprehensive strategy - Sanofi aims to become an AI-driven pharmaceutical leader, with AI embedded in strategic efforts from drug discovery to patient care - Netflix has built its core strategy around its AI recommendation engine for personalization and customer retention	- Alet (2024) - Bahoo et al. (2023) - Brock and Kohli (2023) - Jarrahi et al. (2023) - Lichtenthaler (2018)

Table 12.1. (Continued)

Perspective	Description	Real-Life Examples	Seminal Articles
AI as Strategy	AI is the central focus and driving force of the organization's strategy. The business model and value proposition revolve around developing or deploying AI technologies. AI capabilities are the core differentiator and competitive advantage rather than a supplementary tool.	- Incumbent players across industries are integrating AI into strategies for efficiency, innovation, and competitive advantage - AI-native companies like DeepMind, Anthropic, OpenAI, Grammarly, where the primary offerings are AI technologies and solutions themselves - Surge of generative AI startups in 2023–2024 with business models centred on AI from inception (131 funded by Y Combinator alone) - Tech giants like Google declaring a strategic shift to an 'AI-first' orientation across all business lines	Ruokonen and Ritala (2024) - Industry reports: State of AI 2023 (Benaich, 2023), Y Combinator Startup Directory (2024)

Source: Authors own work.

4.1 AI for Strategy

The perspective of AI for Strategy can be thought of as the first phase in starting to apply AI in a strategic context. In this sense, while AI might not be mentioned explicitly in company strategic initiatives or not play a major role in the overall

strategy, it is utilized to support and drive the organization towards strategic goals. This perspective also entails those companies that are on the periphery of strategic AI utilization (Borges et al., 2021), thus representing the majority at the time of writing (see e.g. Patience & Immerman, 2023).

The use of AI is becoming strategic to companies and industries where it was previously not playing a major role. As in for example, Dzhusupova et al. (2024), present AI as a tool to optimize business operations and decision-making in Engineering, Procurement, and Construction industry. The authors address the strategic choice of the approach towards AI project development, which involves assessing various factors like costs, schedule, business criticality, and IP to make informed decisions. The use of AI is aimed at creating business value by applying it to reduce redundant work and increase efficiency in engineered installations.

While in different industries, the study by Gallego-Gomez and De-Pablos-Herederro (2020) emphasizes AI's role in augmenting the banks' strategic decision-making processes, primarily through better customer relationship management, fraud detection, and enhanced overall service responsiveness. Here too, AI is seen acting as a tool or enabling technology that supports these firms' strategic objectives, such as gaining a competitive advantage or improving operational efficiency. The enhancing role of AI in strategy is implemented rather instrumentally, in a similar fashion as what is discussed by Dzhusupova et al. (2024).

Furthermore, Gudigantala et al. (2023) and Kitsios and Kamariotou (2021) emphasize the importance of aligning AI initiatives with overarching business strategies. Gudigantala et al. (2023) emphasize the importance of a company's business strategy driving AI-driven business outcomes and measurements, which in turn guide AI implementation decisions. As a case, they highlight McDonald's, which focussed on its business strategy to guide AI implementation and pilots (p. 75). AI is thus regarded as a means to achieve strategic goals, rather than an end in itself. The suggestion to use a performance management system like objectives and key results (OKRs) to ensure tight alignment and transparency across the company further solidifies the perspective that AI is being employed for strategic purposes. Thus, Gudigantala et al. (2023) point towards AI being a strategic tool for value creation, much like Kitsios and Kamariotou (2021) who outline that AI can significantly contribute to value creation in organizations through process automation, knowledge acquisition for decision-making, and broadening market opportunities.

Particularly, Kitsios and Kamariotou (2021) discuss AI's role in facilitating the strategic decision-making process by processing vast quantities of data to derive business insights. For instance, applying AI and machine learning within organizations to improve strategic alignment with IT, enhancing decision-making processes, and service innovation reflects AI's enabling capacity for a business's strategic endeavours. These capabilities are not producing opportunities for only large enterprises, but as Shaik et al. (2024) articulate, the capabilities of SMEs can also be increased by AI, for example, when striving for sustainability goals like carbon neutrality. As Goralski and Tan (2020) further discuss, concepts such as sustainable development are often complex issues where AI can aid in supporting

the problem-solving and decision-making that they require. [Watson et al. \(2021\)](#) stress that especially senior leaders should see AI as an important tool to increase their strategic capabilities – acting not just as tools for the company, but even as personalized intelligence agents for the executives.

While the previous viewpoint is argued more from the perspective of giving executives more focussed time on more complex settings, authors such as [Kanitz et al. \(2023\)](#) have already witnessed some capabilities of generative artificial intelligence (GenAI), which can aid directly in the strategizing process. This is a significant shift from the perspective of decision-makers, as only a few years ago many business professionals still perceived AI as primarily used for cost savings and revenue increases ([Güngör, 2020](#)).

Moreover, [Kar et al. \(2021\)](#) propose conceptual frameworks showing hierarchical relationships among various drivers and barriers to AI adoption within organizational strategy. AI's decision-making ability and accuracy are identified as the most influential drivers, suggesting that AI's capability to process and analyze data plays a crucial role in strategic planning and execution.

Together, these perspectives offer insights into the evolving landscape of AI for organizational strategies, enriching our understanding of its applications and strategic implications across different contexts.

4.2 AI in Strategy

The landscape of contemporary business strategy is undergoing a profound transformation with the increasing integration of AI into organizational frameworks. This evolution encompasses a spectrum of approaches, from leveraging AI to enhance existing strengths (AI for strategy) to pioneering new business models and competitive advantages (AI in Strategy). The position of AI in Strategy can be achieved, i.e. transitioning from AI for Strategy, by both looking forward at how AI can best enhance and create value – or, as the Chief Innovation Officer of Philips argues, looking backward by first identifying aspects from business strategy that need support and then align best ways that AI can be utilized ([Ransbotham et al., 2019](#)).

In recent years, there has been a notable surge of Generative AI capabilities, and companies from diverse sectors including big pharma, are shifting towards not only using AI as a tool for supporting, e.g. R&D, but to positioning their organizational strategy to incorporate AI fully. This is exemplified by events such as the statement of the CEO of Sanofi, Paul Hudson: 'Our ambition is to become the first pharma company powered by artificial intelligence at scale' ([Sanofi, 2023](#)). Others include mergers and acquisitions, i.e. BioNTech acquiring InstaDeep, an AI company for €500M ([InstaDeep, 2023](#)), and partnerships such as AstraZeneca and Verge Genomics, with deals worth up to €840M ([Verge Genomics, 2023](#)).

Scholarly discourse further illustrates the multifaceted role of AI within strategic frameworks. E.g. [Lichtenthaler \(2020\)](#) discusses how most firms currently focus heavily on technology issues for their digital transformation initiatives – sometimes at the expense of considering market-related challenges – AI is more

often a crucial element of strategic renewal, aligning with the AI in Strategy perspective where AI is part of a multifaceted strategy. Similarly, [Lichtenthaler \(2018\)](#) proposes the integration of AI into firms' strategies to create new competitive advantages and new business models. In this sense, AI serves as a fundamental component of broader strategic objectives rather than the end goal itself. [Lichtenthaler \(2018\)](#) also presents the concept of 'meta-intelligence', a higher-order intelligence that governs the transformation and alignment of AI and Human Intelligence (HI) with corporate strategies. This is an interesting concept, since the 'transformation of intelligence' is already happening, for example in the corporate innovation domain (e.g. in Pharma-industry; [Kanter, 2024](#)).

The integration of AI into strategy extends to incumbent players, which have had AI as part of their strategies from the beginning, and have obtained their market position largely through leveraging AI. A prime example of this is Netflix, which has had its AI recommendation system at the core of its customer engagement and retention strategies. However, it too faces pressures of the current developments in Generative AI ([Spangler, 2024](#)), where AI might also start to seep into the content its customers consume and what it also produces itself.

[Alet \(2024\)](#) discusses the integration of AI into the business strategy by highlighting four foundational pillars. These include a focussed strategy with AI as a central tool, thorough customer knowledge acquisition, effective customer interaction, and effective AI implementation. The case of Lemonade Insurance exemplifies this perspective by utilizing AI to offer differentiated services, which are part of a broader business model without traditional agents or brokers. AI aids businesses in data collection and interpretation to deliver personalized and real-time services to customers, suggesting how AI tools align with strategic objectives like customer-centricity. The importance of structuring AI projects within competing priorities reflects an integration of AI into an overarching business strategy. [Alet \(2024\)](#) advocates for a synergistic approach where AI becomes part of the strategic fabric of a company.

Furthermore, AI's integration within business strategy extends beyond operational enhancements to encompass innovation research and corporate learning processes. For example, [Mariani et al. \(2023\)](#) highlight AI as a component of business strategy that can often intertwine with core products and operations directly, and especially in the current fast pace of generative AI, it has become extensively connected to innovation research. In this vein, [Bahoo et al. \(2023\)](#) explore AI's multifaceted role in corporate innovation, analyzing how AI technologies are adopted and integrated into various stages of the innovation process, including business model innovation (BMI), product innovation, and innovation outcomes. In this sense, AI is seen as a crucial, integrative part of the firm's broader strategic framework. It is not the central focus but a significant element in achieving strategic objectives. [Bahoo et al. \(2023\)](#) present a framework that reflects the integration of AI in companies' strategies, demonstrating how it aids in decision-making, design thinking, and improving competitiveness.

[Brock and Kohli \(2023\)](#) acknowledge AI as a pivotal component in achieving strategic objectives within the realm of Digital Exploration services. This emerging IT service space goes beyond traditional IT services and digital exploitation services,

by helping organizations develop innovative strategies and processes through advanced digital technologies, such as deep learning and AI. While AI is not the sole focal point, it plays a critical role in enabling new strategies and processes that enhance internal and external effectiveness. Examples provided include the use of AI for market intelligence generation, customer relationship management, and supplier–customer behaviour analysis to guide customers through their journeys. The article demonstrates AI’s embeddedness in modern strategic initiatives. The key points supporting these designations include the integration of AI into developing and enabling new strategies and processes, collaboration between providers and customers on innovation, and the consideration of AI and other digital technologies as foundational elements of new business models.

Jarrahi et al. (2023) argue AI as an integral part of the organizational learning process, suggesting that AI’s power lies in its contribution to strategic learning, capability development, and forming core competencies. The authors do not focus on AI merely as a decision-making tool or the focus of the business strategy itself. Instead, they emphasize its role in augmenting organizational processes, enhancing the knowledge base, and propelling strategic learning, thus reinforcing AI’s place within the broader strategic framework of the organization.

In a guest editorial for the *International Journal of Emerging Markets*, Zhou et al. (2022) showcase via a conceptual framework how AI-driven business strategies are shaped within institutional environments of emerging markets. They argue that businesses must adopt, utilize, integrate, and implement AI solutions as tools to not just enhance productivity but also to lower barriers to doing business, and in the end to foster economic development. Zhou et al. (2022) see businesses needing to conform to social norms and ethical standards when implementing AI, which indicates AI has a multifaceted and large role in organizational strategic considerations.

A fitting example of an AI In Strategy company is Tesla Inc., which uses AI in development, manufacturing, and end-product to enable smart electric vehicles. Their innovation is also AI-focussed, which is used to bring into reality concepts such as self-driving fully autonomous vehicles, and advanced android-robot Optimus (Tesla, 2024). Interestingly, Tesla is currently one of the leading companies in high-performance GPU clusters – which are notably used for high-performance AI computing (Benaich, 2023).

4.3 AI as Strategy

Ruokonen and Ritala (2024) discuss the transformative potential of AI and emphasize the strategic shift that moves AI from being a supportive technology to the centrepiece of business strategy. With reference to Sundar Pichai’s announcement of Google’s shift to an AI-first world, it suggests that while AI was once seen as supplementary, it is becoming a general-purpose technology with measurable benefits across industries. Ruokonen and Ritala (2024) state that there are at the time of writing still relatively few ‘AI-first’ companies. However, AI-first companies are emerging at rapid speed due to advancements in Generative AI – a notable start-up

incubator the Y Combinator has funded 131 Generative AI companies in the year 2023 and the first-quarter of 2024 alone (Y Combinator, 2024).

Companies such as OpenAI present the stereotype of a company that has its hands and feet firmly positioned around artificial intelligence, with the organization practically built upon developing and delivering AI. The mission statement of OpenAI articulates: ‘OpenAI is an AI research and deployment company. Our mission is to ensure that artificial general intelligence benefits all of humanity’ (OpenAI, 2024). AI as Strategy thus highlights those organizations, which have AI-centric approach that fundamentally defines the organization’s value proposition, resource allocation, and innovation processes, as AI becomes the central engine powering growth, efficiency, and differentiation.

A critical aspect of the ‘AI as Strategy’ perspective is the alignment of the organization’s business model and go-to-market strategies around AI. Companies embracing this approach often develop AI-powered products, services, and platforms that form the core of their offerings. They may also leverage AI to transform their internal operations.

5. Discussion

Overall, the reviewed literature underscores the evolving landscape of AI within organizational strategies through AI for Strategy, AI in Strategy, and AI as Strategy. These themes encapsulate varying levels of integration and utilization of AI to drive strategic objectives within organizations. The *AI for Strategy* illustrates organizations leveraging AI as a supportive tool to drive strategic objectives forward (Dzhusupova et al., 2024; Gallego-Gomez & De-Pablos-Heredero, 2020). This illustrates how AI optimizes business operations and decision-making processes, contributing to efficiency and value creation. Furthermore, the alignment of AI initiatives with overarching business strategies highlights AI’s strategic importance in achieving organizational objectives (Gudigantala et al., 2023; Kitsios & Kamariotou, 2021). However, organizations are increasingly integrating AI into strategic frameworks, i.e. *AI in Strategy*, aiming to pioneer new business models and gain competitive advantages. AI is becoming an integral component rather than merely a supportive tool (Alet, 2024; Lichtenthaler, 2018).

Furthermore, a transformative shift was identified where AI emerged as the focal point of organizational strategies. This less studied theme is exemplified by the emergence of ‘AI-first’ companies, where AI is prioritized as the foundation of strategic initiatives, i.e. *AI as strategy* (Ruokonen & Ritala, 2024).

While the ‘Three Perspectives of AI and Strategy Framework’ provides a clear categorization of how organizations integrate AI into their strategic initiatives, it is crucial to understand that these perspectives are not static. Organizations may evolve from one perspective to another as their AI capabilities mature, market dynamics shift, or strategic priorities change. Many organizations might begin their AI journey by leveraging technology as a tool to support and enhance their strategic decision-making processes. This AI for Strategy perspective allows companies to gain familiarity with AI’s potential and build the necessary infrastructure and talent

to support more extensive AI initiatives. However, as these organizations start to realize the tangible benefits of AI and develop a deeper understanding of its capabilities, they may begin to explore ways to integrate AI more deeply into their overall strategic framework.

The transition from AI for Strategy to AI in Strategy is often driven by a recognition that AI can offer more than just decision support. By embedding AI into core business processes, products, and services, organizations can unlock new sources of competitive advantage and drive innovation. This transition requires a significant shift in mindset, as AI moves from being a peripheral tool to a central component of the organization's strategic fabric (refer [Fig. 12.2](#)).

Key drivers of this transition include:

- Proven success of AI initiatives in delivering business value.
- Increasing market pressure to adopt AI to stay competitive.
- Availability of more advanced AI technologies and talent.
- Growing executive support and sponsorship for AI initiatives.

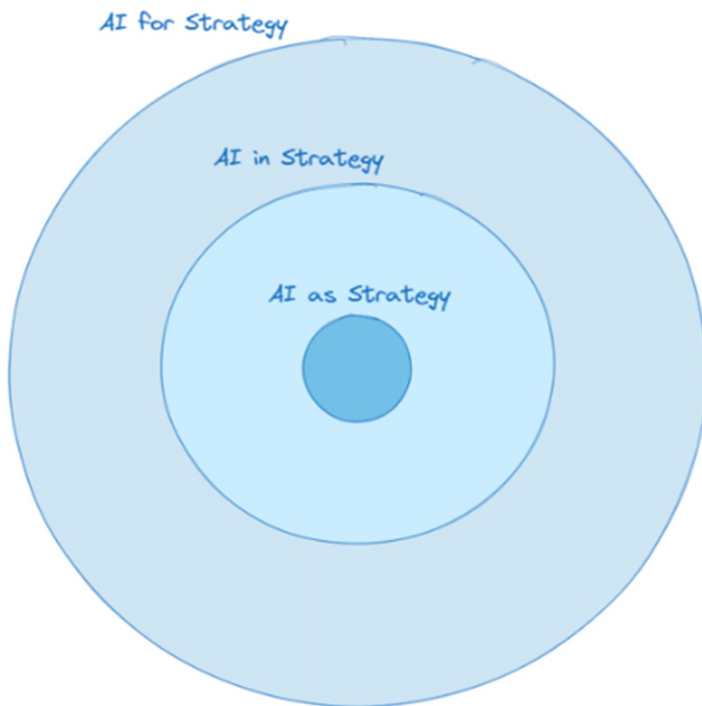


Fig. 12.2. Three Perspectives of AI and Strategy. *Source:* Authors own work.

However, this transition also represents several challenges, such as:

- Aligning initiatives with overall business strategy.
- Developing the necessary talent and infrastructure to support AI at scale.
- Managing the cultural and organizational changes required to embed AI into core processes.
- Addressing ethical and governance concerns around AI's expanded role.

As organizations deepen their AI capabilities and realize the transformative potential of the technology, some may take the bold decision to pivot towards an AI-centric business model. This transition from AI in Strategy to AI as Strategy represents a fundamental shift in the organization's identity and value proposition.

Companies that embrace an AI as Strategy perspective position themselves as leaders in AI innovation, making the technology the core of their competitive advantage. This transition often involves a significant reallocation of resources towards AI research and development, talent acquisition, and infrastructure development. It may also require a reimagining of the organization's business model, partnerships, and go-to-market strategies.

Key drivers of this transition include:

- The emergence of new market opportunities driven by AI.
- Increasing investor and stakeholder interest in AI-centric business models.
- Availability of cutting-edge AI research and talent.
- Vision and leadership of executive champions who believe in AI's transformative potential.

However, the transition to an AI as Strategy perspective also comes with significant risks and challenges, such as:

- High upfront investments in AI research and development with uncertain returns.
- Intense competition for top AI talent and intellectual property.
- Rapidly evolving technological and regulatory landscape that can disrupt AI-centric business models.
- Ethical and societal concerns around the widespread deployment of AI.

While the 'three perspectives on Strategy and AI' has conceptual closeness to frameworks such as the AI Maturity Model (Alsheibani et al., 2019; Sadiq et al., 2021), they differ in several ways. The AI Maturity Model tends to focus on evaluating an organization's internal processes, infrastructure, and overall readiness to adopt AI (e.g. Sadiq et al., 2021) and as such takes an inside-out view for intra-organizational assessment. In contrast, the three perspectives examine how AI strategy aligns with the external business context, taking a market-centric outside-in view. The AI maturity model also has a more operational orientation

geared towards AI implementation – assessing data pipelines, model validation, IT-business alignment, etc. The three perspectives consider strategic dimensions regarding AI's role in business priorities, products/services, and decision ecosystems.

Maturity models enable self-assessment for organizations to identify gaps to improve, while the three perspectives serve more as an analytical lens to understand and compare AI strategic positioning across companies and sectors. Maturity models also imply a linear progression along a spectrum of defined stages towards an 'ideal' end-state (Alsheibani et al., 2019; Sadiq et al., 2021). However, the three perspectives represent distinct strategic choices not bound to a single trajectory.

In essence, the AI Maturity Model takes an inward planning view evaluating organizational readiness while the three perspectives provide an outward market-focussed strategic lens examining AI's role regarding external ecosystem actors. The lenses are therefore complementary for getting a full picture when formulating AI strategy.

This comprehensive literature review and conceptual analysis reveal distinct strategic perspectives in which organizations leverage AI to drive competitive advantage. By categorizing academic discourse and connecting them to real-world cases, three primary viewpoints emerge in how firms align AI capabilities with overarching priorities.

Firstly, some pioneers adopt an 'AI as Strategy' approach, building their entire business model around pioneering AI research and development. This perspective sees AI itself as the core product or service, exemplified by AI-focussed companies like DeepMind, Anthropic, OpenAI, and Grammarly. Here the emphasis is on attracting top talent in AI research, creating novel algorithms and applications, and commercializing technological breakthroughs.

Secondly, many established companies integrate AI tools to optimize operations, customer interactions, and data-driven decision-making. This 'AI in Strategy' approach interweaves AI as part of broader strategic goals like efficiency, personalization, and agility. For instance, retailers may leverage AI for personalized recommendations, while financial institutions apply predictive analytics to guide investments. In these cases, AI enhances, rather than defines, the overarching strategy. In a similar fashion, companies such as Tesla incorporate AI at the heart of their product, and also leverage AI in their manufacturing and R&D processes – AI thus situates deeply in their strategy.

Finally, a prevalent tactical approach utilizes AI more akin to a tool task such as for data processing and analysis to uncover market trends, predict scenarios, and inform planning. This 'AI for Strategy' perspective harnesses AI's analytical prowess to reveal insights and patterns supporting human strategic decision-making, and to enhance operational capabilities. Industries from manufacturing to healthcare are augmenting analysis and planning through this third-party strategic enablement capacity of AI.

Across orientations, the framework provides clarity on AI's evolving role, from bolstering analysis to spearheading innovation. For leaders navigating increasingly AI-driven landscapes, these lenses reflect on current positioning and

paths ahead. As capabilities progress, AI as Strategy may become tomorrow's differentiator, while AI for Strategy sustains near-term gains. Ultimately by aligning AI initiatives with strategic context, firms can unleash this technology's abundant potential.

This conceptual contribution lays the foundation to also track AI's strategic maturation across sectors. Longitudinal data can reveal diffusion patterns of the three perspectives and how organizations transition between them over time. As analytics commoditize and AI-centric business models emerge, the predominant orientation may shift from supplementary to central. Furthermore, fine-grained case analyses can uncover nuances within each perspective regarding types of AI leveraged, implementation challenges, and performance outcomes. In conclusion, by framing the strategic possibilities of AI, this work informs both analytic clarity and foresight for practitioners and scholars navigating the machine age.

Reflecting the intertwining of Strategy and AI could not be timelier, as AI adoption has been largely steady in the few years before the entry of generative AI in 2023 (e.g. [IBM, 2022, 2024](#); [McKinsey & Company, 2022, 2023](#)). From the viewpoint of the three perspectives, it can be speculated that there will be gravitation towards AI as Strategy – companies in the coming years, fuelled by the potential opportunities that generative AI as a general-purpose technology ([Crafts, 2021](#)).

According to a recent report on the State of AI in Nordics by [Silo \(2024\)](#), increases of investment in AI Strategy has increased from 29% in 2022 to 48% in 2023. However, they also report that 'unclear business strategy/roadmap' as the biggest reported challenge in scaling the use of AI across the entire company, rose from less than 2% in 2021 to nearly 17% in 2020 and to a substantial 37% in 2023. This hints at the challenges in transitioning between the baskets of AI and Strategy perspectives, where, i.e. positioning from AI for Strategy into AI in Strategy is difficult since there is a lack of clarity and alignment for enabling the scaled implementation of AI.

Furthermore, [Ransbotham et al. \(2019\)](#) in collaboration with the Boston Consulting Group found that those companies that derive value from utilizing artificial intelligence were more likely to integrate AI with their corporate strategies.

From a practitioner's perspective, the 'Three Perspectives of AI and Strategy Framework' can be a valuable tool for leveraging AI in the context of market intelligence. By understanding how AI can be integrated into strategic initiatives at different levels, organizations can gain a deeper understanding of their markets, customers, and competitive landscape, ultimately driving greater value creation and competitive advantage.

6. Conclusion

Following our literature review analysis, three broad thematic perspectives on AI and Strategy have emerged. It is imperative, however, to delineate the contextual meanings of these perspectives.

The broadness of our chosen thematic perspectives, namely AI for Strategy, AI in Strategy, and AI as Strategy, is justified by the diverse and multifaceted nature of the relationship between AI and organizational strategy. These themes encompass a wide range of subtopics and considerations within the realm of AI integration in strategy. Furthermore, three thematic perspectives in this study differentiate from some of the other popular views of looking at organizations from the viewpoint of AI, such as the AI Maturity Model, which in essence is a tool for intra-organizational inspection of the level of ability of an organization to leverage AI. Instead, the three views of AI and Strategy that we identified can be used as a lens to see how the organizations have their strategies aligned in respective of AI. Thus, the focus is on how AI is currently implemented in strategy, rather than highlighting organizational capacity for implementing AI in a broader context.

Firstly, the theme of AI for strategy refers to the role of AI as a tool for enhancing strategic decision-making processes within organizations. This encompasses various applications of AI, such as data analysis, market trend predictions, and scenario planning, aimed at providing valuable insights to inform strategic decisions. By adopting a broad perspective, we can encompass the myriad ways in which AI facilitates strategic thinking and planning across different industries and organizational contexts. In this view, AI is primarily used as a means to enhance strategic decision-making. This could involve using AI for data analysis, market trend predictions, customer behaviour forecasting, or scenario planning. The idea is to, e.g. leverage AI's ability to process and analyze large datasets more efficiently than humans, providing insights that inform strategic decisions. For instance, a financial institution might use AI to analyze market trends and consumer data to develop investment strategies or to identify emerging market opportunities.

Secondly, the theme of AI in Strategy acknowledges the integration of AI as a crucial element within the broader strategic framework of organizations. This perspective recognizes that AI serves as more than just a standalone tool but is deeply intertwined with other strategic resources and initiatives. By examining AI's role within the broader strategic landscape, we can better understand how organizations leverage AI alongside other strategic assets to achieve their goals and objectives. This perspective integrates AI into the broader strategic framework of an organization, often both in practice and in explicit statements in strategy documentation. It sees AI as a crucial element in achieving strategic goals but not as the sole focus. For example, a media streaming company might incorporate AI recommendation system into the core of its service offering, acting as an integrated component in its entire value offering (i.e. Netflix).

Thirdly, the theme of AI as Strategy acknowledges cases where AI transcends its role as a mere tool or component and becomes the very essence of an organization's strategic direction. This perspective recognizes organizations that prioritize AI as their core business model or value proposition, positioning AI at the forefront of their strategic endeavours. By considering AI as a strategic imperative rather than just a supporting tool, we can explore how organizations redefine their business strategies to capitalize on AI's transformative potential. In this view, AI is not just a tool or a component of strategy, but the strategy itself. Organizations adopting this perspective might focus on AI as their primary business model or

value proposition. For instance, a tech company might centre its entire business strategy around developing cutting-edge AI solutions, positioning AI as its main product or service (i.e. OpenAI). This approach often requires a deep commitment to AI research and development, a focus on attracting top talent in AI, and a business model that revolve around AI technologies.

In summary, the broadness of our chosen thematic perspectives allows for a comprehensive exploration of the diverse ways in which AI intersects with organizational strategy, accommodating various viewpoints, applications, and implications within this dynamic and rapidly evolving field. We fully encourage further research into the intersection of AI and Strategy, especially considering recent technological developments, inside the context of AI as Strategy.

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